The methods text

What the report says in words: the account the tool writes from what was actually done, and the paragraphs the user writes around it.

AT A GLANCE

Where	Summary ⟶ Text
What it is	The prose of the generated report. Two kinds: fixed commentary the tool supplies for each plot, and text blocks written here.
Where it appears	Only in the report built on Summary ⟶ Report. Nothing written here changes a plot or a table.
Numbers	Filtering, normalization, imputation, data source, features used and the significant counts are written by the tool from the session, not typed.
Stored	With the browser session. Kept in a saved state file; lost when the tab is closed without saving.
In viewer mode	This tab is not shown.

1 What the page is for

A report has to say what was done to the data before it says what the data showed, and most of that is already known to the tool: the filter threshold, the normalization, the imputation parameters, the test, the thresholds and the counts they selected. The tool writes those sentences itself, with the numbers filled in from the session. What it cannot know is the experiment — the material, the question, and what a particular figure is being shown for — and that is what this page is for. Text written here is attached to one plot and to one side of it, and appears only in the generated document.

2 Controls

Controls on this page

Control	What it does	Default	Where
Select Plot	Which section of the report the text belongs to. Introduction is the block before the table of contents; every other entry is one plot. Only the plots of the levels actually loaded are offered.	Introduction	page
Position	Whether the text is printed above the figure or below it. Not shown while Introduction is selected.	Above	page
Introduction Text / Text	The text itself. Line breaks are kept; the box is plain text, and anything that looks like markup is printed as characters rather than interpreted.	empty	page
Add / Update	Stores the box under the current plot and position, replacing what was there. A box holding only whitespace stores nothing and clears nothing.	action	page
Delete	Removes the entry for the current plot and position and empties the box.	action	page
Show stored text entries	Prints every stored entry, with its key, so the whole set can be checked at once.	off	page

> NOTE
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> An entry is keyed by the plot *and* the position, so one plot can carry two blocks — one above the figure and one below it — and changing **Position** moves to the other entry rather than moving the text. The editor always shows the entry for the current combination; switching **Select Plot** without pressing **Add / Update** discards what was typed.

Which plots can carry text

The list is exactly the set of sections the report can contain, and no others: coverage, missing values, retention time, modification and missed cleavage at peptide level; coverage, missing values, histogram intensity, boxplot intensity, coefficient of variation, PCA, abundance, correlation and volcano at protein level; those same nine at phosphosite level, with the phosphosite overview added and coverage split into (Number) and (Quality). Only the levels actually loaded are offered. An entry written for a section the report does not end up containing — a volcano text with no comparison in the report, a protein text in a phospho-only session — is kept but never printed.

3 What the tool writes for you

Four things are written without being typed. They appear in this order inside a plot section: the fixed commentary, then the text block for Above, then the figure with its own provenance lines, then the text block for Below.

THE FIXED COMMENTARY

One to four paragraphs under each plot heading, the same words in every report. They describe what the plot is and what a healthy dataset looks like in it; they contain no numbers from the session. They are read from report_general_comments.json beside the application, and re-read every time a report is built, so editing that file changes the commentary of the next report. A plot with no entry in the file gets no commentary paragraph at all. The file is part of the source, so it can be edited by anybody who hosts the application or builds it themselves, and the next report is written with the edited text. It is not editable from the interface.

FILTERING AND NORMALIZATION

Once under each level heading, and again beside every figure that names a data source. Two lines:

- Filtering: either Rows with at least n valid values or Rows with at least p% completeness, followed by the scope — in every condition group, in at least one condition group, or across all annotated samples, independent of condition groups — and were retained (X of Y rows), where Y is the number of rows of the uploaded matrix and X the number that survived the filter. With a threshold of zero, which is the default, the line reads No completeness filter was applied (all Y rows retained).
- Normalization: one of three fixed sentences, for median-centring, quantile normalization (cited as Bolstad et al. 2003) or VSN (cited as Huber et al. 2002). With no normalization the line is left out. All three methods run *before* filtering. Median and quantile methods work on log values; VSN is fitted to linear intensities and returns a log-like analysis scale, and its sentence adds that the CV and abundance plots use the unnormalized data, because VSN output is not on a linear intensity scale.

The whole block is omitted for a level whose annotation carries no filter configuration.

IMPUTATION

Once under each level heading, and again beside every figure that names a data source. With no imputation configured, or with one that was configured and refused, the sentence is that none was applied. Otherwise two sentences. The first is the draw, according to which mode was used: for the default method, a normal distribution centred on the q quantile of the observed values with the adjusted standard deviation times their standard deviation, cited as Lazar et al. 2016; for numerical, a shifted normal distribution stated as a downshift in standard deviations and a width, cited as Tyanova et al. 2016 (the Perseus downshift). The second says whether the parameters were taken sample-wise or across the whole dataset. No random seed is reported.
Beside a figure the wording turns on the figure's own source: a figure whose source is not Imputed Data is described as using the non-imputed data, whatever imputation the level holds.

DATA SOURCE, FEATURES USED, STATISTICS, RESULT

The PCA sections and every volcano comparison carry labelled lines. Data source names the matrix the figure below was drawn from: Data, Filtered Data or Imputed Data. Features used, on the PCA only, reads U of T features (P%) were complete across every sample and entered the projection, and, when $U < T$, says how many were dropped and why. Statistics, on a volcano only, gives the test, the two thresholds and the correction, citing Welch 1947 for the Welch test and Benjamini and Hochberg 1995 for the adjustment; Result, also volcano only, gives how many entities the thresholds selected, named by entity — proteins, phosphosites — rather than *features*.

4 Which number comes from which matrix

This is the part to read before quoting a sentence. The generated prose mixes counts taken from three different matrices, and it labels each one, but the labels are easy to read past.

- *Y, the source rows.* The uploaded matrix, before the $\log_2$ transform and before any filter. It is the only place this number appears.
- *X, the filtered rows.* The matrix after filtering and normalization. It appears in the filtering sentence and nowhere else unless a figure names Filtered Data.
- *Every quality-control figure and its table* — coverage, missing values, histogram, boxplot, coefficient of variation, abundance, correlation, and the peptide and phosphosite plots — is drawn from the matrix *as uploaded,* $\log_2$*-transformed and normalized*: unfiltered, unimputed. Quality control has no data source choice by design, since it exists to describe what the instrument produced. The coefficient-of-variation and abundance plots read that same matrix back on the linear scale, which is the scale those statistics are defined on; after VSN

they read the unnormalized data instead, because VSN output is not on a linear intensity scale.

- *The PCA* is the one exception, and takes the source named on its own line. Its Features used count is smaller again: a projection cannot use a feature with a missing value, and a feature with no variance across the samples contributes nothing, so U counts the complete *and* varying rows of the selected matrix.
- *The volcano counts* come from the matrix named in that comparison's own Data source line, which is the one stored with the comparison when it was added.

EASY TO GET WRONG

The filtering sentence stands above figures that were not filtered. A report can say … were retained (6,214 of 8,806 rows) and, two paragraphs below, draw a coverage plot whose bars count the 8,806. Both are correct and both are labelled; the sentence describes what filtering did to the level, not what the figure under it shows. When a count is quoted in a paper, quote it with the matrix it came from.

What the volcano result line counts

HOW IT IS COMPUTED

The comparison is condition$_1$ − condition$_2$: a feature called *upregulated* is higher in the first condition named, and the sentence says so in words as well as by the label.

- *Tested* is not the number of rows in the matrix. A row enters the test only if it has at least two non-missing values in *each* of the two conditions, and, for a paired test, at least two usable pairs. Rows below that are dropped before the test and are in none of the counts.
- *Upregulated* is adjusted p below the threshold *and* $\log_2$ FC above the fold-change threshold; *downregulated* is adjusted p below the threshold and $\log_2$ FC below minus the threshold. Both comparisons are strict.
- *Not significant* is everything else *except* rows without a p-value, which are classed separately and counted in none of the three. A row gets no p-value when its p-value is not finite or when it has no variation (every compared condition constant; for a paired test, identical differences). When there are such rows the line ends with their count, *k* without a p-value, so the four counts add up to the tested total.
- With **Use uncorrected p-values** on, the threshold is applied to the raw p-value and the line says p rather than adjusted p.

FIXED PROSE, NOT GENERATED

The volcano commentary gives an adjusted p-value of 0.05 and an absolute $\log_2$ fold change of 1 as commonly used examples, not as a claim about the current comparison.

> The generated Statistics line states the actual test, correction and thresholds. Quote that line when describing the analysis, and check any text you added yourself against it.

5 What the text cannot say

> **NOT A METHODS SECTION**
>
> What the tool writes is an account of the processing, not of the experiment. Nothing in it names the instrument, the gradient, the search engine, the database or its version, the enzyme, the modifications, the FDR applied during the search, the number of biological replicates or what they are replicates of. None of that reaches the tool, so none of it can be written; a methods section built on the generated text alone is a methods section missing its first half. The Introduction block is the place to put it.

Edge cases

- A text block is stored under the plot's internal key. Unloading a level removes its plots from **Select Plot**, but the blocks written for them are kept and reappear when the level is loaded again.
- Text entries survive a save and restore of the session state. They are not cleared when a new dataset is uploaded — unlike the logged volcano comparisons, which are.
- If the comments file cannot be read when the report is built, the report is still built, with the commentary from the last successful read. The warning is shown in the **Report Warnings** panel on Summary → Report only; warnings never appear in the report.

Behind this page

Function	What it produces
report_function	Assembles the report, and with it every sentence described here
_filter_description	The filtering, row-count and normalization sentences
_imputation_paragraphs	The imputation sentences, shared by the level preamble and every figure that names a source
_data_provenance_html	The Data source, Filtering, Imputation and Features used lines
_pca_completeness_description	The count of features that entered the projection
_volcano_statistics_html	The test, thresholds and correction line
_volcano_counts_html	The result line, counted with the same routine the statistics page reports